Crossed recoupling in the local dS_3 model: physical normalization and a Euclidean check

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Abstract

We match the scalar channels reached by two ordered local creations to standard conformal partial waves. Three-point group normalization and an independent partial-wave norm fix the conversion to the known conformal $6j$ symbol, including its phase. A direct Euclidean area integral checks the complex matrix element independently of its Lorentzian inversion formula. We give a conservative analytic bound on the three puncture regions and formulate the crossed overlap for physical creation states. The calculation concerns a declared $G_N = 0$ model. The scalar block is a compression of the full Racah transform; it is not claimed to be unitary. The full physical wave-packet integral remains to be evaluated.

1 Model, channels, and normalization

Set the de Sitter radius to one, $G_N = 0$, and use the proper group $SO^+(3,1)$. Two scalar Bunch–Davies observers have labels at least $s_* > 0$. The distinct free matter species have positive labels p, q; local couplings are one. Matter modes have Klein–Gordon normalization, and observer modes have the previously fixed spacetime delta normalization. Write

$$h_{\mathrm{m}}(x) = \sqrt{1+x^2}, \qquad \rho(x) = 2x^2/\pi, \qquad \mu_{\mathrm{obs}}(x) = x\tanh(\pi x)/\pi.$$

The compact subgroup has Haar probability, with radial push-forward $\sinh^2 r\ \mathrm{d}r$. For $S = x+y+z$, retain

$$D_{xyz} = \cosh(\pi S/2) \prod_{n=x,y,z} \cosh[\pi(S-2n)/2], \qquad K_{xyz} = \frac{\pi}{8D_{xyz}} \prod_{n=x,y,z} \frac{\sinh(\pi n)}{n}.$$

These conventions and the physical four-body norms were derived in our preceding two-matter note. They are inputs here.

Fix the external order $e = (a, q, c, p)$. The s pairing is $(a, q)|(c, p)$, while the LPRS t pairing is $(c, q)|(a, p)$ [1]. The latter is the reverse of the pair order naturally used by the opposite creation sequence. This reversal must be accompanied by an internal shadow before normalization. The full CG decomposition includes all integer principal-series spins [3, 2]. Each local scalar observer creation reaches spin zero in its own pairing. This does not imply that the scalar sector is complete after recoupling. We calculate the 00 matrix element of the full transform.

2 From sphere periods to physical channel states

Let $\Delta_x = 1 + ix$. Denote the all-ket boundary trilinear's spherical 000 component by $B(x, y, z)$. For $\nu_i = (x, y, z)$, put

$$\alpha_i = -\frac{1}{2} - \frac{i}{2}(S - 2\nu_i), \quad A = \sum_i \alpha_i, \qquad B = (4\pi)^{3/2} 4^A \frac{\Gamma(A+2)\prod_i \Gamma(\alpha_i+1)}{\prod_i \Gamma(A-\alpha_i+2)}.$$

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Its modulus is

$$|B(x,y,z)|^2 = \frac{\pi^4}{D_{xyz}} \prod_{n=x,y,z} \frac{\sinh(\pi n)}{n}. \tag{1}$$

Negative labels in B denote shadow representatives, not negative masses. Use standard shadow-integral partial waves

$$\Psi^s_b = \int \mathrm{d}^2x\, t_{\Delta_a\Delta_q\Delta_b}\, t_{\widetilde{\Delta}_b\Delta_c\Delta_p}, \qquad \Psi^t_d = \int \mathrm{d}^2x\, t_{\Delta_c\Delta_q\Delta_d}\, t_{\widetilde{\Delta}_d\Delta_a\Delta_p}.$$

Projection of two external spherical $l = 0$ states leaves only internal $l = 0$. The remaining normalized spherical harmonics pair to one. Consequently the 0000 components are exactly

$$N_s(b) = B(a,q,b)B(c,p,-b), \qquad N_t(d) = B(c,q,d)B(a,p,-d). \tag{2}$$

The plane-to-sphere unitary conformal map introduces no further constant: the two contracted dimensions sum to two and absorb its area Jacobian. In the unique scalar CG channel, the $0000 = 1$ states therefore are

$$J_s(b) = \Psi^s_b/N_s(b), \qquad J_t(d) = \Psi^t_d/N_t(d). \tag{3}$$

For conformal weights $h = (1+ib)/2$, $h_i = (1+ie_i)/2$, use

$$K^{ij}_h = \pi \frac{\Gamma(2h-1)}{\Gamma(2-2h)} \frac{\Gamma(1-h-h_i+h_j)\Gamma(1-h+h_i-h_j)}{\Gamma(h-h_i+h_j)\Gamma(h+h_i-h_j)}.$$

The Gamma identities in B and the two shadow-block coefficients give

$$\frac{N_s(b)}{N_s(-b)} = \frac{K^{34}_{1-h}}{K^{12}_{1-h}}, \qquad \frac{\Psi^s_b}{N_s(b)} = \frac{\Psi^s_{-b}}{N_s(-b)}. \tag{4}$$

Thus the raw waves need not be equal under reflection. Exchanging the two trilinears also reflects the internal label; after division by N, it gives the same J. This connects the LPRS t orientation to the actual $(a,p)|(c,q)$ creation channel.

3 The physical crossed kernel

KKS fixes the scalar three-point quotient pairing to $[2^d \operatorname{vol} SO(d-1)]^{-1}$, hence $1/4$ at $d = 2$ [2, (13)–(15)]. In our group convention,

$$\|t_{xyz}\|^2_{\mathrm{phys}} = |B(x,y,z)|^2/K_{xyz} = 8\pi^3.$$

The common Haar-volume ratio between our physical quotient and the LPRS/KKS quotient is therefore $32\pi^3$. It is fixed by the three-point pairing, not fitted to a four-point value. The previously derived scalar four-body rigging weights are

$$W_s(b) = \rho(b)K_{abq}K_{bcp}, \qquad W_t(d) = \rho(d)K_{cdq}K_{dap}.$$

A distribution $\int F(b)J_s(b)\,\mathrm{d}b$ has physical norm $\int |F(b)|^2/W_s(b)\,\mathrm{d}b$. Equation (1) gives

$$\frac{|N_s(b)|^2}{W_s(b)} = \frac{32\pi^7}{b^2}. \tag{5}$$

Independently, the scalar shadow square is $\mathcal{N}_b = \pi^2/b^2$. KKS (74) gives $\mu_{\mathrm{KKS}} = b^2/(8\pi^3)$, distinct from μ_{obs}. Their (39) yields, with positive-label measure $\mathrm{d}b$,

$$n_b = (1/4)^2\, 2\pi/\mu_{\mathrm{KKS}} = \pi^4/b^2.$$

This agrees with (5) divided by $32\pi^3$. One reflection representative has already been selected; no extra factor of two is present. Let $\mathcal{J}_2(b,d;e) = \langle \Psi^s_b, \Psi^t_d \rangle_{\mathrm{LPRS}}$ be the known scalar conformal $6j$ symbol [1, (3.4)–(3.5)]. For delta-normalized states $e^s_b = \sqrt{W_s(b)}J_s(b)$,

$$\langle J_s(b), J_t(d) \rangle_{\mathrm{phys}} = \frac{32\pi^3 \mathcal{J}_2(b,d;e)}{\overline{N_s(b)} N_t(d)}, \tag{6}$$

$$U_{00}(b,d;e) = \frac{bd}{\pi^4} \frac{N_s(b)/|N_s(b)|}{N_t(d)/|N_t(d)|} \mathcal{J}_2(b,d;e). \tag{7}$$

The phases cannot be replaced by modulus squares. This dictionary follows the actual local CG chains. It does not undo the previously found normalization-invariant mismatch of the separate H/Z candidate kernels.

4 An independent Euclidean overlap

Gauge-fix $(z_1, z_2, z_3, z_4) = (0, z, 1, \infty)$ and define

$$k_h(z) = z^h \, {}_2F_1(h - h_{12}, h + h_{34}; 2h; z).$$

For scalar principal series, $\bar{h} = h$, not h^*. The full wave is

$$F_s(z) = |z|^{-2(h_1+h_2)} \left[K^{34}_{1-h} k_h(z) k_h(\bar{z}) + K^{12}_h k_{1-h}(z) k_{1-h}(\bar{z}) \right]. \tag{8}$$

For F_t, replace the external order by $(3,2,1,4)$, the internal label by d, and z by $1-z$. The direct check is

$$\mathcal{J}_2 = \frac{1}{4} \int_{\mathbb{C}} \mathrm{d}^2 z \, \overline{F_s(z)} F_t(z). \tag{9}$$

This never calls the Lorentzian inversion's ${}_4F_3$ or Euler-product evaluator. It shares only the standard three-point shadow coefficients. Use the base domain $|z| < 1$, $|1-z| > 1$ and its six images

$$z, \quad 1-z, \quad 1/z, \quad 1/(1-z), \quad z/(z-1), \quad (z-1)/z.$$

They cover the plane once away from boundaries: their domains correspond to the six orderings of $|z|, |1-z|, 1$. Their area Jacobians are $1, 1, |z|^{-4}, |1-z|^{-4}, |1-z|^{-4}, |z|^{-4}$. On the upper half put $z = re^{i\theta}$ with $\arccos(r/2) < \theta < \pi$. Scalar waves are unchanged by interchanging $z, \bar{z}$, so the lower half supplies a factor two. The combined quotient factor is $1/2$. We use $r = t^P$, $P = 4$ or 8, and Gauss–Legendre nodes in t and the mapped angle, retaining every complex chart contribution.

5 A conservative analytic puncture bound

The shadow integral has real part one in every dimension. Hence

$$|F_s(z)| \le \frac{I(z)}{|z|}, \quad |F_t(z)| \le \frac{I(z)}{|1-z|}, \qquad I(z) = \int_{\mathbb{C}} \frac{\mathrm{d}^2 x}{|x||x-z||x-1|}.$$

For $r = |z| \le 1/4$, split the x plane at radii $2r, 1/2, 2$. The contributions obey 40π, $8\pi \log(1/(4r))$, 48π, $16\pi/7$, respectively, so

$$I(z) \le 8\pi[12 + \log(1/r)]. \tag{10}$$

For the first region set $x = ry$ and use $|y| + |y - z/r| \ge 1$ to bound the reciprocal product by $2(|y|^{-1} + |y - z/r|^{-1})$. Integrating over disks of radii two and three gives 40π after bounding $|x-1|^{-1} \le 2$. In the middle annulus the integrand is at most $4/|x|^2$. In the third it is at most $8/|x-1|$ on a disk of radius three; in the last it is at most $16/(7|x|^3)$.

Reflection $z \mapsto 1 - z$ controls the second puncture. Scaling $x = zy$ gives $I(z) = |z|^{-1} I(1/z)$ for the third. If all six charts omit $r < \epsilon$, their images lie in disks of radius $\delta = \epsilon/(1-\epsilon)$ around zero and one, and outside $|z| = 1/\delta$. For $\delta \le 1/4$, put $V = 12 + \log(1/\delta)$ and $C = 128\pi^3/3$. Integration gives the raw $6j$ tail bound

$$T(\epsilon) = C \left[2\delta(V^2 + 2V + 2) + \frac{\delta^2}{2} \left(V^2 + V + \frac{1}{2} \right) \right]. \tag{11}$$

This proves absolute convergence of the crossed overlap and removal of the puncture cutoff. It is deliberately conservative and is not a Gauss quadrature error bound. The same-channel delta norm is a different, distributional question.

6 Physical ordered states and the next integral

Let the initial vacuum coefficient be $f(c)$ with $\|f\|_0^2 = \int \rho(c) |f(c)|^2 \, \mathrm{d}c$. Write E_{xyk} for the previously fixed real local 000 emission amplitude. At real times, the ordered channel amplitudes are

$$v_s(a,b,c) = \frac{f(c) E_{abq} E_{bcp}}{\sqrt{W_s(b)}} \, e^{-it_p[h_{\mathrm{m}}(b) - h_{\mathrm{m}}(c)] - it_q[h_{\mathrm{m}}(a) - h_{\mathrm{m}}(b)]}, \tag{12}$$

$$v_t(a,d,c) = \frac{f(c) E_{adp} E_{dcq}}{\sqrt{W_t(d)}} \, e^{-iu_q[h_{\mathrm{m}}(d) - h_{\mathrm{m}}(c)] - iu_p[h_{\mathrm{m}}(a) - h_{\mathrm{m}}(d)]}. \tag{13}$$

All observer labels are at least s_*. Their crossed overlap is

$$X = \int \mathrm{d}a \; \mathrm{d}c \; \mathrm{d}b \; \mathrm{d}d \, \overline{v_s(a,b,c)} \, U_{00}(b,d;e) \, v_t(a,d,c), \tag{14}$$

understood as the Hilbert-space limit of compact spectral restrictions, without assuming absolute convergence of the fourfold integral. The previously proved second-mass-moment condition $\int \rho(c) |f(c)|^2 c^2 \, \mathrm{d}c < \infty$ is sufficient for the unregulated two-step states. Shifting both creations by the proper-mass half period multiplies either amplitude by $\exp\{\pi[h_{\mathrm{m}}(a) - h_{\mathrm{m}}(c)]/2\}$; those blocks are bounded on every initial L^2 state.

The full Racah transform is unitary, so its scalar compression is a contraction. Thus $|X| \le \|v_s\| \|v_t\|$. If compact restrictions give $v_{s,L}, v_{t,L}$, then

$$|X - X_L| \le \|v_s - v_{s,L}\| \|v_t\| + \|v_{s,L}\| \|v_t - v_{t,L}\|. \tag{15}$$

These tails are positive norms, computed from the known density $\rho(c)|f(c)|^2 R_q(a|b) R_p(b|c)$ and its exchanged counterpart (or the half-period rates $\widetilde{R}$). This provides a route to a controlled physical packet calculation without asserting scalar unitarity.

7 Evidence and limits

Three label sets pass 33 normalization, phase, channel-exchange, and shadow-reflection checks at 45 decimal digits. For

$$e = (0.4, 0.7, 0.93, 0.6), \quad b = 0.65, \quad d = 0.57,$$

the raw and physical matrix elements are

$$\mathcal{J}_2 \simeq 190.9971396918 + 3.7321669491i, \qquad U_{00} \simeq 0.7266051675149.$$

The complex Euclidean relative differences at orders $8, 16, 24, 40$ with $P = 4$ are approximately $1.13\,10^{-2}$, $8.93\,10^{-5}$, $5.13\,10^{-6}$, and $1.01\,10^{-7}$. With $P = 8$, orders 24 and 40 give $2.81\,10^{-7}$ and

$8.73\,10^{-11}$. Stored data retain imaginary parts, chart contributions, absolute contribution sums, grids, precision, and dependency versions. For $e = (1.1, 0.43, 0.82, 1.3)$, $b = 0.37$, $d = 1.17$, the $P = 8$ relative differences at orders 24 and 40 are $3.37\,10^{-5}$ and $4.58\,10^{-8}$. The absolute-sum cancellation ratios are about 2.17 and 3.66 in the two final grids. At the first point, cutting the chart radius at 10^{-3} and 10^{-4} gives relative differences $5.06\,10^{-4}$ and $9.56\,10^{-5}$; these include quadrature error and are not isolated tail measurements. The analytic tail bound at $\epsilon = 10^{-12}$ is less than $4.38\,10^{-6}$ in absolute raw-kernel units. Ten further diagnostics check the chart coverage, the known area integral $\frac{1}{4}\int(1+|z|^2)^{-2}\,\mathrm{d}^2z = \pi/4$, and the tail primitive. The area relative error at 32 nodes is below $1.3\,10^{-40}$. An initial source comment misstated the base domain; its angular implementation was correct. The executed source is retained, and an AST comparison confirms that only the comment changed. These are empirical checks, not interval certificates or uniform statements over the spectrum.

The new work is the physical normalization dictionary, its coordinate check, and the explicit domain and cutoff formulation. The conformal $6j$, CG completeness, and shadow coefficients have the cited predecessors. The actual fourfold physical packet integral, identical-species Bose interference, full Fock domains, finite-G_N dynamics, and a complete de Sitter holographic dual remain open.

References

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